

Fall 2025 Econ 191 paper: Measuring the Impact of Israel’s 2021–2023 Port Reforms on Labor Productivity Mediated by Capital Deepening

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Abstract

This paper evaluates the impact of Israel’s recent (2021–2023) port reforms on the sector’s persistently low labor productivity and examines the role of capital deepening in mediating these effects. The reforms were aimed at breaking monopoly conditions by introducing intra-port competition through new deep-water terminals in Haifa and Ashdod, and by privatizing Haifa’s legacy operator. Besides the overall effect of the reforms on labor productivity, the study quantifies the share of productivity gains associated with capital deepening, a channel of special interest given the uniquely (among comparable OECD members) low levels of capital per worker in Israel’s non-tradable sectors. I use event-study difference-in-differences estimators for staggered adoption to evaluate the impact of each of the three reforms on output per worker and capital per worker at the port/terminal-month level. I then estimate the elasticity of labor productivity with respect to capital–labor ratios in a fixed-effects panel framework and combine these estimates with the event-study coefficients to decompose the reform-induced productivity gains into a component attributed to capital deepening and a residual component. I find that

1 Introduction

2 Literature Review

3 Data and Empirical Strategy

3.1 Overview and raw data

The empirical analysis combines administrative throughput and service-quality data for Israel’s two main container ports, Haifa and Ashdod, with firm-level financial statements and project-level information for Haifa Port Company. From the port side, I assemble a port/terminal–month panel of container throughput (TEU), total cargo tonnage, and container-handling productivity (TEU per work-hour) published by the Administration of Shipping and Ports. These series provide the raw quantity and performance inputs for constructing output, labor hours, and labor productivity at the

terminal-month level. On the capital side, I use Haifa Port Company’s audited financial statements (balance sheet PPE positions, Note 8 roll-forwards, and fixed-asset cash-flow items), together with national capital-goods deflators and a project-level dataset of major investments and disposals, to build a monthly real capital stock for the legacy Haifa container operations.

Table 1 summarizes the observed throughput and TEU-per-hour KPI series that underlie the output and labor measures. Table 2 summarizes the raw capital inputs—net PPE stocks, annual flows, and the main dated investment and disposal events—that are later transformed into a monthly K series using a perpetual-inventory-type procedure. The remainder of this section describes these data sources in more detail and explains how they are combined to construct the variables used in the empirical models.

Table 1: Observed throughput series and extracted KPIs (coverage and counts)

Port/Terminal	Variable	Granularity	Range	N
Haifa	Throughput (TEU)	Monthly	01-2008–10-2021	166
Haifa	Throughput (tons)	Monthly	01-2008–08-2025	200
Ashdod	Throughput (TEU)	Monthly	01-2008–10-2021	166
Ashdod	Throughput (tons)	Monthly	01-2008–08-2025	200
Haifa–Bayport	Throughput (TEU)	Quarterly	Q3-2021–Q4-2024	14
Haifa–Bayport	Throughput (tons)	Monthly	07-2021–08-2025	50
Ashdod–HCT	Throughput (TEU)	Quarterly	Q3-2021–Q4-2024	14
Ashdod–HCT	Throughput (tons)	Monthly	08-2021–08-2025	49
Haifa	KPI: TEU/hour (Π)	Annual	2018–2024	7
Ashdod	KPI: TEU/hour (Π)	Annual	2018–2024	7
Haifa–Legacy	KPI: TEU/hour (Π)	Annual	2018–2024	7
Ashdod–Legacy	KPI: TEU/hour (Π)	Annual	2018–2024	7
Haifa–Bayport	KPI: TEU/hour (Π)	Annual	2021–2024	4
Ashdod–HCT	KPI: TEU/hour (Π)	Annual	2022–2024	3
Haifa, Ashdod, terminals	Labor hours (proxy, L)	Monthly (terminal)	2018-01–2024-12	N

3.2 Data sources

This subsection briefly inventories the main data sources before turning to the construction of the key variables in Section ?? . I distinguish between throughput and operational data, labor inputs, capital and project-level data, and time-varying controls.

Throughput and operational data. Monthly port-level TEU and tonnage series for Haifa and Ashdod, and quarterly terminal-level TEU for Bayport and HCT, are drawn from the Administration of Shipping and Ports’ statistical publications. These form the basis for the port/terminal-month output measure used in the labor-productivity calculations. The same sources provide annual terminal-level TEU-per-hour KPIs and, in some cases, waiting, berth, and total stay times per ship call; the latter are used as auxiliary outcomes and controls.

Table 2: Raw capital inputs for Haifa legacy port (PPE stocks, flows, and major projects)

<i>Panel A: Haifa Port Company net PPE and annual flows (Track B)</i>			
Year	Net PPE (kNIS, EoY)	Additions to PPE (kNIS)	Depreciation expense (kNIS)
2018	1,163,539	30,896	1,132,643
2019	1,010,641	25,025	985,616
2023	1,044,165	95,565	948,600
2024	966,652	57,732	908,920
<i>Panel B: Major capital projects and asset transactions (Haifa legacy)</i>			
Project	Area/terminal	Event type	Commissioning
Carmel container terminal (Phase I)	Carmel container terminal	Addition	2010-09 / 2011-03
Tugboat “Ilan 1”	Marine fleet	Addition	2011-03
Tugboat “Elad”	Marine fleet	Addition	2011-03
Sea department transfer to IPC	Sea department	Disposal	2020-01
Kishon West rehabilitation (HPC share)	Kishon West terminal	Addition	2025-12
Mobile crane and heavy equipment	Kishon pier / projects areas	Addition	2026-06
Five new general-cargo cranes	Kishon West terminal	Addition	2023-09 / 2024-03
Waterfront relocation: new port infra	New operations areas	Addition	2024-03

Labor data. Direct monthly labor-hours by terminal are not available in published form. For the current version of the analysis, I therefore construct a terminal-month labor input series $L_{i,t}$ by inverting the TEU-per-hour KPIs and allocating implied annual hours across months in proportion to each terminal’s within-year TEU profile. This proxy preserves the annual identity between TEU, hours, and TEU-per-hour and treats pre-opening periods as structural zeros. Once administrative monthly labor-hours are obtained from Israel Ports Company and the legacy operators, the same pipeline will be rerun with true $L_{i,t}$, leaving the empirical design unchanged but improving measurement of both $LP_{i,t}$ and $K/L_{i,t}$.

Capital and project-level data. The capital measurement relies on Haifa Port Company’s annual financial statements, including balance sheet net PPE, Note 8 fixed-asset roll-forwards (gross cost, accumulated depreciation, additions, disposals), and fixed-asset items in the statement of cash flows, as summarized in Panel A of Table 2. These nominal series are combined with a national capital-goods deflator to obtain a real net PPE series. Panel B of Table 2 lists the major dated investment and disposal events used to overlay lumpy shocks on the PPE-based capital path: the Carmel container terminal expansion, new tugboat acquisitions, the 2020 sea-department transfer, Kishon West crane and infrastructure projects, and the waterfront relocation program. The detailed algorithm for turning these inputs into a monthly K_t series is described in Section ??.

Time controls and other covariates. In addition to the core port/terminal variables, the analysis incorporates time-varying controls that capture macro and sector-specific shocks, including COVID-19 disruptions, major security events, and calendar seasonality. These are implemented as calendar month-year fixed effects in the baseline specifications, with additional event-specific dummies used in robustness checks where appropriate.

3.3 Construction of key variables

This subsection defines the main quantities used in the empirical analysis and explains how the raw series described above are transformed into the panel used for estimation. I first define output and labor productivity at the terminal-month level, then describe the construction and status of the labor input series, introduce the capital stock and capital-labor ratios used as mediators, and finally set out the structure of the estimation panels and sample restrictions.

3.3.1 Output and measurement of labor productivity (LP)

The primary unit of observation in the analysis is a container terminal i in calendar month t . For each port $p \in \{\text{Haifa, Ashdod}\}$ and month t , the Administration of Shipping and Ports reports total container throughput in TEU, $\text{TEU}_{p,t}$, and total cargo throughput in metric tons, $\text{Tons}_{p,t}$ (Table 1). Once the new deep-water terminals enter, terminal-level TEU is observed at quarterly frequency, $\text{TEU}_{i,q}$, where q indexes calendar quarters.

To obtain a terminal-month output measure, I allocate port-level monthly TEU to terminals using observed quarterly shares. Let $I(p)$ denote the set of terminals operating in port p in quarter q , and define the quarterly share of terminal $i \in I(p)$ in port p as

$$s_{i,p,q} = \frac{\text{TEU}_{i,q}}{\sum_{j \in I(p)} \text{TEU}_{j,q}}.$$

For each month t in quarter $q(t)$, terminal-level TEU is then defined as

$$\widehat{\text{TEU}}_{i,t} = s_{i,p(i),q(t)} \cdot \text{TEU}_{p(i),t},$$

where $p(i)$ denotes the port of terminal i . Before a new terminal opens, its share is structurally zero and $\widehat{\text{TEU}}_{i,t} = 0$.

Because container terminals handle heterogeneous cargoes (e.g. mixed container weights, different cargo types), I work with a mix-adjusted output measure that scales TEU by a port-month specific tons-per-TEU weight. For each port p and month t with positive TEU and tonnage, define

$$w_{p,t} = \frac{\text{Tons}_{p,t}}{\text{TEU}_{p,t}},$$

and winsorize $w_{p,t}$ at suitable lower and upper percentiles to limit the influence of outliers. Terminal-month output is then defined as

$$Y_{i,t} = w_{p(i),t} \cdot \widehat{\text{TEU}}_{i,t},$$

which can be interpreted as a tons-equivalent measure of container throughput at terminal i in month t . This mix-adjusted $Y_{i,t}$ is the baseline output concept used in the labor productivity

calculations.¹

Given output $Y_{i,t}$ and a measure of labor input $L_{i,t}$ (described in the next subsection), labor productivity at the terminal-month level is defined as

$$LP_{i,t} = \frac{Y_{i,t}}{L_{i,t}},$$

and the empirical analysis works with its logarithm,

$$\ell p_{i,t} = \log LP_{i,t} = \log Y_{i,t} - \log L_{i,t}.$$

The canonical analysis file used in the event-study regressions stacks these terminal-month observations over time and across Haifa–Legacy, Haifa–Bayport, Ashdod–Legacy, and Ashdod–HCT, applying a simple source-precedence rule: when both port-level and terminal-level throughput data are available, the terminal-level values are taken as primary and port-level series are used only to backfill missing months via the allocation procedure above.

The main estimation sample for labor productivity is a balanced window from January 2018 through December 2024, during which both throughput and TEU-per-hour KPIs are available for the legacy terminals and the new entrants. Months in which both $Y_{i,t}$ and $L_{i,t}$ are structurally zero (e.g. pre-opening months for new terminals) are excluded from regressions; months with $Y_{i,t} > 0$ but mechanically imputed hours are retained but flagged for robustness checks.

3.3.2 Labor input: from proxy L to true L

Direct monthly labor-hours by terminal are not available in published form. In the current version of the analysis, I therefore use a proxy labor input constructed from the TEU-per-hour KPIs and throughput series. Let $\Pi_{i,y}$ denote the annual TEU-per-hour KPI for terminal i in year y , and let $\widehat{\text{TEU}}_{i,y}$ denote terminal-year TEU aggregated from the monthly series $\widehat{\text{TEU}}_{i,t}$. The implied annual hours for terminal i in year y are

$$H_{i,y} = \frac{\widehat{\text{TEU}}_{i,y}}{\Pi_{i,y}}.$$

These implied annual hours preserve by construction the identity $\widehat{\text{TEU}}_{i,y} = \Pi_{i,y} \cdot H_{i,y}$ at the terminal-year level.

To obtain a monthly series, I distribute $H_{i,y}$ across months within the same year in proportion to each terminal’s monthly TEU profile:

$$L_{i,t}^{\text{proxy}} = H_{i,y(t)} \cdot \frac{\widehat{\text{TEU}}_{i,t}}{\sum_{s \in \mathcal{M}(y(t))} \widehat{\text{TEU}}_{i,s}},$$

where $y(t)$ denotes the calendar year of month t and $\mathcal{M}(y)$ is the set of months in year y . For months in which $\widehat{\text{TEU}}_{i,t} = 0$ for all t in year y , the terminal is treated as inactive and $L_{i,t}^{\text{proxy}}$ is

¹In robustness checks, I also consider specifications that use $\widehat{\text{TEU}}_{i,t}$ directly as output, without mix adjustment.

set to zero (structural zero). This construction ensures that terminal-year hours sum to $H_{i,y}$, that terminal-month hours track observed within-year variations in throughput, and that pre-opening periods for new terminals do not contribute spurious labor inputs.

The proxy $L_{i,t}^{\text{proxy}}$ is used to construct baseline labor productivity $LP_{i,t} = Y_{i,t}/L_{i,t}^{\text{proxy}}$ in the current version of the paper and is the labor input entering the event-study models in Section ?? . Its validity is assessed using a set of QA checks (comparing implied hours to reported staffing and shift patterns where available, and checking for implausible month-to-month fluctuations in hours). However, the proxy has clear limitations: it is constructed from the same KPIs that enter the numerator of productivity and may smooth away intensive-margin labor adjustments.

For the final version of the thesis, the goal is to replace this proxy with administrative monthly labor-hours by terminal obtained from Israel Ports Company and the legacy operators. The empirical design is constructed so that this replacement is mechanical: once monthly hours $L_{i,t}^{\text{true}}$ are available, they will simply be substituted for $L_{i,t}^{\text{proxy}}$ in the definitions of $LP_{i,t}$ and $K/L_{i,t}$, leaving the regression specifications and identification strategy unchanged. In that final setup, the proxy-based results will be retained for comparison and robustness, with the main tables and figures based on $L_{i,t}^{\text{true}}$.

3.3.3 Capital stock and capital-to-labor ratios

The capital side of the analysis uses the Haifa Port Company (HPC) financial statements and the project-level data summarized in Table 2 to construct a real capital stock series for the Haifa legacy container operations. Let $\widetilde{K}_y$ denote the end-of-year net PPE for HPC in year y , measured in thousands of NIS and deflated using a capital-goods price index so that $\widetilde{K}_y$ is in constant prices. The annual PPE roll-forwards provide information on additions, disposals, and depreciation expense, which are used to check the consistency of the deflated series and to distinguish between background investment and lumpy project-related changes.

To obtain a monthly capital path for the Haifa legacy container operations, I follow a two-step “Path B” procedure. First, I interpolate between annual real net PPE levels to obtain a smooth background capital process, allocating the net effect of additions and disposals evenly across months within each year. Second, I superimpose discrete jumps at the commissioning months of major projects and asset transactions identified in Panel B of Table 2: the initial Carmel container terminal expansion, new tugboat acquisitions, the 2020 sale of the sea department to Israel Ports Company, the Kishon West crane and rehabilitation projects, and the waterfront relocation investments. For each such event, I assign a real capital increment (or decrement) equal to the project’s real cost (or the book value of disposed assets, where available) and apply it at the recorded commissioning month. The result is a monthly real capital stock series

$$K_t^{\text{Haifa-legacy}}, \quad t = \text{month},$$

for the Haifa legacy container operations.

At this stage, the detailed Path B series is implemented for the Haifa legacy operator. For

terminals without a dedicated capital series (e.g. Bayport, HCT, and Ashdod legacy), $K_{i,t}$ is not yet observed and $K/L_{i,t}$ is therefore not used in the baseline regressions. The K/L analysis and mediation results presented in Section ?? are thus based on the Haifa legacy terminal sample, with extensions to other terminals left for future work as additional capital data become available.

Given a capital stock series $K_{i,t}$ and labor input $L_{i,t}$, the capital–labor ratio at terminal i in month t is

$$\frac{K_{i,t}}{L_{i,t}},$$

and the empirical analysis works with the log capital–labor ratio

$$c_{i,t} = \log \left(\frac{K_{i,t}}{L_{i,t}} \right) = \log K_{i,t} - \log L_{i,t}.$$

For the Haifa legacy container terminal, $K_{i,t} \equiv K_t^{\text{Haifa-legacy}}$ as constructed above, and $L_{i,t}$ is either the proxy or the true monthly hours once available. The log capital–labor ratio $c_{i,t}$ plays two roles in the empirical strategy: it is an outcome in its own right in the event-study analysis (Model 1B), and it serves as the key explanatory variable in the elasticity model (Model 2) used to decompose labor-productivity responses into a component consistent with capital deepening and a residual.

3.3.4 Final analysis panels and sample restrictions

The variables defined above are assembled into a set of analysis-ready panels tailored to the different empirical models in the paper.

For the main event-study models (Models 1A and 1B), the estimation sample is a terminal–month panel covering Haifa–Legacy, Haifa–Bayport, Ashdod–Legacy, and Ashdod–HCT from January 2018 through December 2024. The outcome for Model 1A is $\ell p_{i,t} = \log LP_{i,t}$; for Model 1B, it is the log capital–labor ratio $c_{i,t}$, currently available for Haifa–Legacy. Each reform is associated with an event-time clock and the not-yet-treated (NYT) design described in Section ?. Observations for which both $Y_{i,t}$ and $L_{i,t}$ are structurally zero (pre-opening months) are excluded; observations with implausible productivity values (e.g. extremely low $L_{i,t}$ combined with positive $Y_{i,t}$) may be trimmed based on pre-specified percentile rules. Months heavily affected by the COVID-19 pandemic and by major security shocks are retained in the baseline but flagged; robustness checks will show results excluding those periods.

For the elasticity and mediation model (Model 2), the estimation sample is restricted to terminal–month observations for which both $\ell p_{i,t}$ and $c_{i,t}$ are observed and reliably measured. In the current implementation, this implies a focus on the Haifa legacy container terminal over the same 2018–2024 window, with the possibility of incorporating additional terminals once comparable capital series are constructed. The panel used for Model 2 is identical in structure to that for Model 1A, but with the additional requirement that $K_{i,t}$ is observed.

In some cases, it is convenient to work with aggregated panels—for example, port–quarter aggregates or stacked event-study datasets in which each reform is represented as a separate pseudo-

panel. These derived panels are constructed mechanically from the underlying terminal-month data (by summing or averaging within port-quarter cells and by stacking reform-specific event windows) and are used only for presentation and estimation convenience. All such transformations are recorded in the analysis code and can be traced back to the canonical terminal-month panel described in this subsection.

4 Econometric Design

This section sets out the empirical strategy for estimating the effects of the port reforms on labor productivity and capital deepening, and for relating these responses to one another. First, I lay out event-study difference-in-differences (DiD) specifications for $\ln(LP_{i,t})$ (Model 1A) and $\ln(K_{i,t}/L_{i,t})$ (Model 1B), using a not-yet-treated (NYT) design in the spirit of Callaway and Sant’Anna (2) and Sun and Abraham (10). I then present a fixed-effects panel regression of $\ln(LP_{i,t})$ on $\ln(K_{i,t}/L_{i,t})$ (Model 2), which provides an elasticity used in a mediation-style decomposition of the reform effects. Throughout, I discuss the key identifying assumptions and potential threats, and how these are addressed in the empirical implementation.

4.1 Event-study DiD for labor productivity and capital-labor ratios (Model 1)

I treat each reform as an event at the terminal-month level:

- Haifa competition entry: introduction of the private Bayport terminal (09-2021);
- Ashdod competition entry: introduction of the private HCT terminal (11-2022);
- Haifa privatization: sale of the Haifa legacy operator via concession (01-2023).

For each reform r and the terminals it affects, I define an event-time variable $M_{i,t}^r$ measuring the number of months relative to the implementation date of that reform for terminal i : $M_{i,t}^r = 0$ in the month when reform r goes live for terminal i , $M_{i,t}^r = -1, -2, \dots$ in pre-reform months, and $M_{i,t}^r = 1, 2, \dots$ in post-reform months. terminal-months that have not yet or never will undergo a reform of the same kind as reform r (competition entry or privatization) are flagged/enter $M_{i,t}^r$ as controls. The not-yet-treated (NYT) design described below determines which terminal-months are handled as treatment and which as control. For a given reform r , I estimate a pair of event-study regressions with the same specification and sample but different outcomes. For labor productivity,

$$\ln(LP_{i,t}) = \sum_{m \in \mathcal{M}_r, m \neq -1} \beta_m^{LP,r} \mathbf{1}\{M_{i,t}^r = m\} + \gamma_i + \delta_t + X_{i,t}'\theta + \varepsilon_{i,t}^{LP,r}, \quad (1)$$

and for the capital-labor ratio,

$$\ln\left(\frac{K_{i,t}}{L_{i,t}}\right) = \sum_{m \in \mathcal{M}_r, m \neq -1} \beta_m^{K/L,r} \mathbf{1}\{M_{i,t}^r = m\} + \gamma_i + \delta_t + X_{i,t}'\theta + \varepsilon_{i,t}^{K/L,r}, \quad (2)$$

where i indexes terminals (or ports in periods before competition entry), t indexes calendar months, γ_i are terminal fixed effects, δ_t are calendar-month fixed effects, and $X_{i,t}$ collects additional controls (e.g. port-specific linear trends or crisis dummies). $\mathcal{M}_r$ denotes the set of months m (leads and lags around reform r) that enter the regression as dummy variables, with the reference month $m = -1$ excluded. That way, each $\beta_m^{LP,r}$ and $\beta_m^{K/L,r}$ measures the change in $\ln(LP)$ or $\ln(K/L)$ at event time m relative to the last pre-reform month, netting out terminal and time fixed effects and controls.

Not-yet-treated design and sample construction. As Sun and Abraham (10) emphasize, in settings with staggered adoption and heterogeneous effects across cohorts and over time, a naive two-way fixed-effects event study implicitly uses already-treated months as controls for later-treated units, which can produce non-transparent and even biased estimates. To circumvent this problem, I implement a *not-yet-treated* (NYT) strategy in the spirit of Callaway and Sant’Anna (2); Sun and Abraham (10). For each reform r and treated terminal, I define treatment at the month when that reform first becomes active for that terminal (event time $M_{i,t}^r = 0$), and then restrict the control months so that treated observations are compared only to terminal-months that have not yet experienced a reform of the same type as r (competition entry or privatization) at month t , or never experience such a reform. More concretely, for each reform r and its corresponding impacted terminals, the sample of (i, t) observations entering (1) and (2) is restricted so that terminal-months with $M_{i,t}^r = m$ (for $m \in \mathcal{M}_r$) are compared only to terminal-months that are not yet treated under a reform of that type at month t ; already-treated terminal-months are excluded from the relevant control group.

Table 3 summarizes, for each reform and regression target, the terminal-months used as treatment and as NYT controls. The same NYT windows are used in the $\ln(LP)$ and $\ln(K/L)$ regressions; only the dependent variable differs.

Table 3: Not-yet-treated windows by reform and regression target

Reform	Regression target	Treatment months	Control months
Haifa competition entry (Bayport opens 09-2021)	Haifa-Bayport terminal	Haifa port: 09-2019 to 08-2021	Ashdod port: 09-2019 to 07-2021
		Haifa-Bayport: 09-2021 to 10-2022	Ashdod-Legacy: 08-2021 to 10-2022 Ashdod-HCT: 08-2021 to 10-2022
	Haifa-Legacy terminal	Haifa port: 09-2019 to 08-2021	Ashdod port: 09-2019 to 07-2021
		Haifa-Legacy: 09-2021 to 10-2022	Ashdod-Legacy: 08-2021 to 10-2022 Ashdod-HCT: 08-2021 to 10-2022
Ashdod competition entry (HCT effective 11-2022)	Ashdod-HCT terminal	Ashdod port: 11-2020 to 07-2021	Haifa port: 11-2020 to 08-2021
		Ashdod-HCT: 08-2021 to 09-2023	Haifa-Bayport: 09-2021 to 09-2023 Haifa-Legacy: 09-2021 to 09-2023
	Ashdod-Legacy terminal	Ashdod port: 11-2020 to 07-2021	Haifa port: 11-2020 to 08-2021
		Ashdod-Legacy: 08-2021 to 09-2023	Haifa-Bayport: 09-2021 to 09-2023 Haifa-Legacy: 09-2021 to 09-2023
Haifa privatization (Haifa-Legacy sold 01-2023)	Haifa-Legacy terminal	Haifa port: 01-2021 to 08-2021	Haifa-Bayport: 09-2021 to 09-2023
		Haifa-Legacy: 09-2021 to 09-2023	Ashdod port: 01-2021 to 07-2021
			Ashdod-Legacy: 08-2021 to 09-2023 Ashdod-HCT: 08-2021 to 09-2023

Under the identification assumptions of the NYT design, the event-time coefficients $\beta_m^{LP,r}$ in (1) and $\beta_m^{K/L,r}$ in (2) can be interpreted as causal effects of reform r on $\ln(LP)$ and $\ln(K/L)$ at event time m , relative to the last pre-reform month. The key assumption is that, in the absence of reform r , treated terminals and the corresponding NYT comparison group would have followed similar trends in labor productivity and in the capital-labor ratio over the estimation window. In Section 5, I will assess this assumption using pre-trend patterns and placebo reform dates.

Dynamic effects and window averages. Running the Model 1 regressions yields dynamic treatment-effect estimates for each reform clock. Unlike settings suited to regression-discontinuity designs, where the parameter of interest is the jump exactly at the cutoff, I expect competition entry and privatization to affect labor productivity and the capital-labor ratio gradually over many months after implementation. Focusing on a single event-time coefficient would therefore miss most of the adjustment path. Instead, I treat the post-reform event-study coefficients $\{\beta_m^{LP,r} : m > 0\}$ and $\{\beta_m^{K/L,r} : m > 0\}$ as dynamic paths of the average treatment effect for reform r .

Following the modern DiD event-study literature (e.g. 2; 10), I summarize these paths by taking simple averages over pre-specified event-time windows. For any window $[a, b]$, I define

$$\bar{\beta}_{[a,b]}^{LP,r} = \frac{1}{b-a+1} \sum_{m=a}^b \beta_m^{LP,r}, \quad \bar{\beta}_{[a,b]}^{K/L,r} = \frac{1}{b-a+1} \sum_{m=a}^b \beta_m^{K/L,r}. \quad (3)$$

Each window estimate is a linear combination of the underlying event-time coefficients and can be

interpreted as the average causal effect of reform r on $\ln(LP_{i,t})$ or $\ln(K_{i,t}/L_{i,t})$ over that window.

In the results tables, for each regression I report five window averages: (i) a “full post” window $m \in [1, M_{\text{post}}]$, where M_{post} is the last observed post-reform month, summarizing the average effect over the entire post period; (ii) a common two-year window $m \in [1, 24]$, which allows direct comparison across ports and reforms; (iii) a first post year $m \in [1, 12]$, capturing short-run effects; (iv) a second post year $m \in [13, 24]$, capturing whether effects persist, strengthen, or decay; and (v) an “average pre” window $m \in [-13, -2]$, which serves as a pre-trend check. Under the identifying assumptions of the NYT design, this average-pre window should be close to zero and statistically insignificant. Window averages are converted to percentage changes in labor productivity and in the capital-labor ratio via $100 \cdot (e^{\bar{\beta}_{[a,b]}} - 1)$

All Model 1 regressions are estimated by ordinary least squares on the terminal-month panel described in Section ??, with standard errors clustered at the terminal level. I report both unweighted and TEU-weighted versions, the latter using terminal-month output $Y_{i,t}$ as regression weights.

4.2 Elasticity of log labor productivity with respect to log capital per worker (Model 2)

Models (1)–(2) characterize how the reforms affect log labor productivity and log capital per worker separately. To relate these two sets of responses, I estimate the elasticity of labor productivity with respect to capital per worker in a fixed-effects panel regression. The baseline specification is

$$\ln(LP_{i,t}) = \alpha_i + \tau_t + \eta \ln\left(\frac{K_{i,t}}{L_{i,t}}\right) + Z'_{i,t}\lambda + u_{i,t}, \quad (4)$$

where α_i are terminal fixed effects, τ_t are time fixed effects, and $Z_{i,t}$ is a vector of additional controls (which may include reform dummies or event-time indicators to ensure that the entire reform effect is not mechanically attributed to $\ln(K_{i,t}/L_{i,t})$). The coefficient η is interpreted as the elasticity of output per worker with respect to capital per worker, conditional on fixed effects and controls. In a Cobb–Douglas benchmark with constant returns to scale, η would coincide with the capital share; here it is a reduced-form elasticity capturing within-terminal co-movements of $\ln(LP_{i,t})$ and $\ln(K_{i,t}/L_{i,t})$ after controlling for common shocks and time-invariant heterogeneity.

The baseline specification uses contemporaneous $\ln(K_{i,t}/L_{i,t})$, but I will report robustness checks using lagged capital-labor ratios (e.g. $\ln(K_{i,t-1}/L_{i,t-1})$) to reduce simultaneity concerns. Estimation is based on the subset of terminal-month observations for which both $\ln(LP_{i,t})$ and $\ln(K_{i,t}/L_{i,t})$ are observed and reliably measured (initially, the Haifa legacy container terminal over 2018–2024). As in Model 1, standard errors are clustered at the terminal level, and TEU-weighted versions of (4) will be reported as robustness checks.

Identification in Model 2 relies on the assumption that, conditional on terminal and time fixed effects and on $Z_{i,t}$, variation in $\ln(K_{i,t}/L_{i,t})$ is sufficiently exogenous with respect to transitory shocks to $\ln(LP_{i,t})$ for the purposes of an accounting-style exercise. Reverse causality and omitted time-varying shocks remain concerns, which motivates the use of lags, crisis controls, and compar-

isons of the estimated η to capital–income shares from Haifa Port Company’s financial accounts. Throughout, I treat η as an interpretive elasticity rather than a fully identified structural parameter.

4.3 Mediation–style decomposition of reform effects

Finally, I combine the event–study estimates from Model 1 with the elasticity estimate $\hat{\eta}$ from Model 2 to construct a mediation–style decomposition of the reform effects on log labor productivity. The goal is to quantify how much of the observed change in $\ln(LP_{i,t})$ is *consistent* with the estimated elasticity and the observed change in $\ln(K_{i,t}/L_{i,t})$, without claiming a fully identified causal mediation parameter.

For each reform r and event–time horizon $[a, b]$, define the total effect of reform r on log labor productivity over that horizon as the window average

$$TE_{[a,b]}^r = \bar{\beta}_{[a,b]}^{LP,r}, \quad (5)$$

and the corresponding average change in log capital per worker as

$$\Delta C_{[a,b]}^r = \bar{\beta}_{[a,b]}^{K/L,r}. \quad (6)$$

Given the estimated elasticity $\hat{\eta}$ from (4), the component of the productivity response that is consistent with capital deepening over horizon $[a, b]$ is approximated by

$$ME_{[a,b]}^r \approx \hat{\eta} \cdot \Delta C_{[a,b]}^r, \quad (7)$$

and the residual component is

$$RE_{[a,b]}^r = TE_{[a,b]}^r - ME_{[a,b]}^r. \quad (8)$$

Whenever $TE_{[a,b]}^r \neq 0$, I also report the share of the total effect accounted for by capital deepening:

$$s_{[a,b]}^r = \frac{ME_{[a,b]}^r}{TE_{[a,b]}^r}. \quad (9)$$

This decomposition is best viewed as a structured accounting exercise. It does not rely on an external instrument for $\ln(K_{i,t}/L_{i,t})$ and does not deliver a formal IV–based mediation effect. Instead, it uses the within–terminal elasticity estimated in Model 2 together with the reform–induced changes in $\ln(K_{i,t}/L_{i,t})$ from Model 1 to quantify how much of the event–study response of labor productivity is consistent with capital deepening, and how much remains to be explained by other channels such as improved utilization of existing capital, changes in work organization, or pure efficiency gains.

5 Results and discussion

This section presents the empirical results from the econometric framework in Section ???. I begin with descriptive evidence on throughput, labor productivity, and capital per worker. I then report not-yet-treated (NYT) event-study estimates for log labor productivity (Model 1A) and log capital per worker (Model 1B), followed by fixed-effects elasticity estimates linking log LP and log(K/L) (Model 2). Finally, I combine these ingredients in a mediation-style accounting exercise that gauges how much of the reform-induced productivity changes can be quantitatively attributed to capital deepening.

Throughout this section, I focus on window-average summaries of the event-study paths, deferring full dynamic coefficients to an appendix.

5.1 Descriptive patterns in throughput, labor productivity, and capital deepening

This subsection will present baseline descriptive evidence on the evolution of throughput, labor productivity, and capital per worker before and after the reforms. A figure analogous to Figure 1 in the proposal will plot $\log LP_{i,t}$ by port and terminal over 2018–2025, with vertical lines marking Bayport’s opening (09/2021), HCT’s opening (11/2022), and Haifa–Legacy’s privatization (01/2023). A companion table will report summary statistics for the main variables used in the regressions (throughput, LP , K , K/L), by port, terminal, and broad pre/post-reform periods.

The text here will briefly describe the main visual patterns (e.g., trends, breaks, and comparisons across terminals), without making causal claims, and will set the stage for the formal event-study estimates in the next subsections.

5.2 Competition entry and labor productivity (Model 1A)

This subsection reports NYT event-study results for the competition-entry reforms, using $\log LP_{i,t}$ as the outcome and the specifications in Section ???. The main estimates are summarized in Table 4, which reuses and updates the proposal layout to focus on window-average effects for Haifa and Ashdod, separately for entrant and legacy terminals.

The text in this subsection will interpret the signs and magnitudes of the window-average effects in Table 4, compare entrant vs. legacy terminals, and use the pre-trend diagnostics as an informal test of the NYT parallel-trends assumption.

5.3 Haifa privatization and labor productivity (Model 1A)

Next, I report NYT event-study results for Haifa–Legacy’s privatization, again using $\log LP_{i,t}$ as the outcome. Table 5 reuses the proposal’s layout, with Panel A reporting the privatization effect on Haifa–Legacy and Panel B showing a placebo application of the same clock to Haifa–Bayport.

The text here will compare the Haifa–Legacy privatization effects to the competition-entry effects, and use the Bayport placebo as an internal falsification check.

Table 4: Average Post-Reform Effects on $\ln(\text{LP})$ by Port and Terminal

Panel A: Haifa (competition clock)				
	<i>SIPG (entrant)</i>		<i>Legacy terminal</i>	
	(1) Baseline	(2) +PortTr	(3) Baseline	(4) +PortTr
All post $m \in [1, 24]$	β (SE)	β (SE)	β (SE)	β (SE)
Implied % Δ LP, $[1, 24]$	$\Delta\%$	$\Delta\%$	$\pm\Delta\%$	$\pm\Delta\%$
Post year 1, $m \in [1, 12]$	β (SE)	β (SE)	β (SE)	β (SE)
Implied % Δ LP, $[1, 12]$	$\Delta\%$	$\Delta\%$	$\pm\Delta\%$	$\pm\Delta\%$
Post year 2, $m \in [13, 24]$	β (SE)	β (SE)	β (SE)	β (SE)
Implied % Δ LP, $[13, 24]$	$\Delta\%$	$\Delta\%$	$\pm\Delta\%$	$\pm\Delta\%$
Average pre, $m \in [-13, -2]$	β (SE)	β (SE)	β (SE)	β (SE)
Pre-trends: F-test p -value	p	p	p	p
Observations	N	N	N	N
Within R^2	R^2	R^2	R^2	R^2

Panel B: Ashdod (competition clock)				
	<i>HCT (entrant)</i>		<i>Legacy terminal</i>	
	(5) Baseline	(6) +PortTr	(7) Baseline	(8) +PortTr
All post $m \in [1, 24]$	β (SE)	β (SE)	β (SE)	β (SE)
Implied % Δ LP, $[1, 24]$	$\Delta\%$	$\Delta\%$	$\pm\Delta\%$	$\pm\Delta\%$
Post year 1, $m \in [1, 12]$	β (SE)	β (SE)	β (SE)	β (SE)
Implied % Δ LP, $[1, 12]$	$\Delta\%$	$\Delta\%$	$\pm\Delta\%$	$\pm\Delta\%$
Post year 2, $m \in [13, 24]$	β (SE)	β (SE)	β (SE)	β (SE)
Implied % Δ LP, $[13, 24]$	$\Delta\%$	$\Delta\%$	$\pm\Delta\%$	$\pm\Delta\%$
Average pre, $m \in [-13, -2]$	β (SE)	β (SE)	β (SE)	β (SE)
Pre-trends: F-test p -value	p	p	p	p
Observations	N	N	N	N
Within R^2	R^2	R^2	R^2	R^2

Outcome is $\ln(\text{LP}_{it})$ at the terminal $\times$ month level. All columns are NYT event-study regressions with terminal and calendar-month fixed effects; columns labeled “+PortTr” additionally include port-specific linear time trends. Event time m is measured in months relative to competition entry, with $m = -1$ omitted; not-yet-treated observations under the relevant competition clock serve as controls. “Implied % Δ LP” converts window averages via $100 \cdot (e^{\bar{\beta}_{[a,b]}} - 1)$. “Average pre, $m \in [-13, -2]$ ” summarizes lead coefficients and is used as a pre-trend check. “Pre-trends: p (Leads F-test)” reports the p -value of a joint test that all lead coefficients $m \leq -2$ are zero. Standard errors (in parentheses) will be clustered by port (wild-cluster bootstrap p -values); within R^2 is the within-variation R^2 from the fixed-effects regression.

Table 5: Haifa Privatization — Dynamic and Window-Average Effects on $\ln(\text{LP})$

	Panel A: Haifa–Legacy (privatization)		Panel B: Haifa Bayport (placebo)	
	(1) Baseline	(2) +PortTr	(3) Baseline	(4) +PortTr
All post $m \in [1, 24]$	β (SE)	β (SE)	β (SE)	β (SE)
Implied % Δ LP, $[1, 24]$	$\pm\Delta\%$	$\pm\Delta\%$	$\pm\Delta\%$	$\pm\Delta\%$
Post year 1, $m \in [1, 12]$	β (SE)	β (SE)	β (SE)	β (SE)
Implied % Δ LP, $[1, 12]$	$\pm\Delta\%$	$\pm\Delta\%$	$\pm\Delta\%$	$\pm\Delta\%$
Post year 2, $m \in [13, 24]$	β (SE)	β (SE)	β (SE)	β (SE)
Implied % Δ LP, $[13, 24]$	$\pm\Delta\%$	$\pm\Delta\%$	$\pm\Delta\%$	$\pm\Delta\%$
Average pre, $m \in [-13, -2]$	β (SE)	β (SE)	β (SE)	β (SE)
Pre-trends: F-test p -value	p	p	p	p
Observations	N	N	N	N
Within R^2	R^2	R^2	R^2	R^2

Outcome is $\ln(\text{LP}_{it})$ at the terminal $\times$ month level. All columns are NYT event–study regressions with terminal and calendar–month fixed effects; columns labeled “+PortTr” additionally include port–specific linear time trends. Event time m is defined relative to the privatization of the Haifa–Legacy terminal (January 2023, $m = 0$); $m = -1$ is omitted, and not–yet–treated observations under the privatization clock serve as controls. Window–average rows report equal–weight means of β_m over the indicated ranges. “Average pre, $m \in [-13, -2]$ ” summarizes lead coefficients and is used as a pre–trend check. “Implied % Δ LP” converts window averages via $100 \cdot (e^{\bar{\beta}_{[a,b]}} - 1)$. “Pre–trends: F-test p -value” reports the p -value of a joint test that all lead coefficients $m \in [-13, -2]$ are zero. Panel A reports privatization effects for Haifa–Legacy; Panel B applies the same clock as a placebo to Haifa–Bayport (SIPG), which is never privatized. Standard errors (in parentheses) will be clustered by port; within R^2 is the within–variation R^2 .

5.4 Capital deepening and elasticity of $\ln(\text{LP})$ with respect to $\ln(K/L)$

I now turn to the capital side of the analysis. Given current data availability, Models 1B and 2 will initially be estimated for Haifa, focusing on Haifa–Legacy (and Bayport as a placebo where appropriate).

Event–study results for $\ln(K/L)$ (Model 1B)

A table parallel to Table 5 will summarize NYT event–study effects of the reforms on $\log(K/L)_{i,t}$ for Haifa, using the same windows and specifications as for $\log \text{LP}_{i,t}$. A template is given below.

Elasticity of $\ln(\text{LP})$ with respect to $\ln(K/L)$ (Model 2)

Table 7 will report estimates of the elasticity η from Model 2 in Equation (4), under several alternative K and L definitions (baseline Path B capital, alternative depreciation rates, alternative labor measures, and a first–difference specification).

The accompanying text will describe the range of estimated elasticities, discuss their plausibility relative to capital–income shares, and note how sensitive $\hat{\eta}$ is to alternative K and L definitions.

5.5 Mediation-style decomposition of reform effects

Finally, I combine the event–study window–average estimates and the elasticity $\hat{\eta}$ to construct the mediation–style decomposition described in Section ?? . Table 8 will summarize, for each reform

Table 6: Haifa — Average Post-Reform Effects on $\ln(K/L)$ (Model 1B)

	(1) Haifa–Legacy	(2) Haifa Bayport (placebo)
All post $m \in [1, 24]$	β (SE)	β (SE)
Implied % $\Delta K/L$, $[1, 24]$	$\Delta\%$	$\Delta\%$
Post year 1, $m \in [1, 12]$	β (SE)	β (SE)
Implied % $\Delta K/L$, $[1, 12]$	$\Delta\%$	$\Delta\%$
Post year 2, $m \in [13, 24]$	β (SE)	β (SE)
Implied % $\Delta K/L$, $[13, 24]$	$\Delta\%$	$\Delta\%$
Average pre, $m \in [-13, -2]$	β (SE)	β (SE)
Pre-trends: F-test p -value	p	p
Observations	N	N
Within R^2	R^2	R^2

Outcome is $\ln(K_{it}/L_{it})$ at the terminal $\times$ month level. Specifications mirror those in Tables 4 and 5, with $\log(K/L)$ as the dependent variable. Window–average rows and implied percentage changes are defined analogously, and pre–trend tests again summarize the lead coefficients. Initial estimates will be reported for Haifa–Legacy (and Bayport as a placebo) over the 2018–2024 period for which K_{it} is reliably constructed.

Table 7: Elasticity of $\ln(LP)$ with Respect to $\ln(K/L)$ (Model 2)

	(1) Baseline FE	(2) Alt K	(3) Alt L	(4) First–diff
$\ln(K/L)_{it}$	β (SE)	β (SE)	β (SE)	β (SE)
Additional controls Z_{it}	Yes	Yes	Yes	Yes
Terminal FE	Yes	Yes	Yes	–
Month FE	Yes	Yes	Yes	–
Terminal $\times$ month FE	–	–	–	Yes
Observations	N	N	N	N
Within R^2	R^2	R^2	R^2	R^2

Outcome is $\ln(LP_{it})$ for terminals with reliable K_{it} and L_{it} . Column (1) reports the baseline fixed-effects estimate of η using Path B capital and the preferred labor–hours series. Columns (2) and (3) replace K and L with alternative constructions (e.g. different depreciation or labor allocation rules). Column (4) estimates a first–difference version of Model 2 as a growth–accounting robustness check. All specifications cluster standard errors at the terminal level. Estimates of η will be compared to benchmark capital–income shares from Haifa’s accounts in the discussion.

r and horizon $[a, b]$, the total effect on $\log LP$, the corresponding effect on $\log(K/L)$, the implied mediated component, and the residual component.

Table 8: Mediation-Style Decomposition of Reform Effects on $\ln(LP)$

Reform r	Horizon $[a, b]$	$TE_{[a,b]}^r$ (total)	$\Delta C_{[a,b]}^r$ (K/L)	$ME_{[a,b]}^r$ (mediated)	$RE_{[a,b]}^r$ (residual)	$s_{[a,b]}^r$ (share)
Haifa competition	[1, 24]	β (SE)	β (SE)	β (SE)	β (SE)	s
	[1, 12]	β (SE)	β (SE)	β (SE)	β (SE)	s
	[13, 24]	β (SE)	β (SE)	β (SE)	β (SE)	s
Ashdod competition	[1, 24]	β (SE)	β (SE)	β (SE)	β (SE)	s
	[1, 12]	β (SE)	β (SE)	β (SE)	β (SE)	s
	[13, 24]	β (SE)	β (SE)	β (SE)	β (SE)	s
Haifa privatization	[1, 24]	β (SE)	β (SE)	β (SE)	β (SE)	s
	[1, 12]	β (SE)	β (SE)	β (SE)	β (SE)	s
	[13, 24]	β (SE)	β (SE)	β (SE)	β (SE)	s

$TE_{[a,b]}^r$ is the window-average effect of reform r on $\ln(LP)$ from Model 1A; $\Delta C_{[a,b]}^r$ is the corresponding window-average effect on $\ln(K/L)$ from Model 1B. The mediated component is $ME_{[a,b]}^r \approx \hat{\eta} \Delta C_{[a,b]}^r$, where $\hat{\eta}$ is the elasticity estimated in Table 7. The residual component is $RE_{[a,b]}^r = TE_{[a,b]}^r - ME_{[a,b]}^r$, and $s_{[a,b]}^r = ME_{[a,b]}^r / TE_{[a,b]}^r$ (when $TE_{[a,b]}^r \neq 0$) is the share of the total effect that is quantitatively consistent with capital deepening. The discussion will emphasize that this is an accounting exercise rather than a formally identified IV-based mediation model.

The narrative in this subsection will highlight, for each reform and horizon, whether capital deepening can plausibly explain most of the observed productivity gains, or whether large residual components point to additional mechanisms such as improved utilization of existing capital, organizational changes, or pure efficiency improvements.

5.6 Robustness and sensitivity analyses (overview)

Detailed robustness checks—including alternative event-time binning, different fixed-effects and clustering schemes, alternative capital and labor series, stacked event-studies, and placebo tests—will be reported in a separate subsection or appendix. Here I will briefly summarize which variants leave the main conclusions unchanged and flag any specifications that materially alter the magnitude or interpretation of the estimated effects.

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Appendix A. Full Dynamic Event-Time Tables

A.1. Full Event-Time Estimates — Haifa

Table 9: Full Event-Time Estimates — Haifa

m	<i>SIPG (entrant)</i>			<i>Legacy</i>			$N(m)$
	(1) Baseline	(2) +PortTr	(3) +Tr&Shocks	(4) Baseline	(5) +PortTr	(6) +Tr&Shocks	
-4	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
-3	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
-2	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
0	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
1	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
2	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
3	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
4	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
5	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
6	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
7	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
8	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
9	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
10	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
11	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
12	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
13	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
14	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
15	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
16	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
17	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
18	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
19	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
20	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
21	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
22	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
23	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
24	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n

Notes: Event time m is measured in months relative to the relevant competition-entry reform ($m = 0$). The month $m = -1$ is omitted as the reference bin. All specifications include terminal and month fixed effects; “+PortTr” adds port-specific linear trends; “+Tr&Shocks” additionally includes COVID (2020–21) and late-2023 war-shock controls. β_m are event-time coefficients with standard errors in parentheses. $N(m)$ reports the number of terminal $\times$ month observations supporting each event-month m under the baseline NYT specification.

A.2. Full Event-Time Estimates — Ashdod

Table 10: Full Event-Time Estimates — Ashdod

m	<i>HCT (entrant)</i>			<i>Legacy</i>			$N(m)$
	(1) Baseline	(2) +PortTr	(3) +Tr&Shocks	(4) Baseline	(5) +PortTr	(6) +Tr&Shocks	
-4	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
-3	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
-2	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
0	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
1	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
2	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
3	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
4	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
5	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
6	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
7	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
8	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
9	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
10	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
11	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
12	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
13	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
14	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
15	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
16	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
17	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
18	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
19	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
20	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
21	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
22	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
23	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n
24	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	n

Notes: Same NYT event-study specification as Table 9, but with event time m defined relative to Ashdod’s competition-entry reform. The omitted reference bin is $m = -1$. All specifications include terminal and month fixed effects; “+PortTr” adds port-specific linear trends; “+Tr&Shocks” additionally includes COVID (2020–21) and late-2023 war shock controls. β_m are event-time coefficients with standard errors in parentheses. $N(m)$ reports the number of terminal×month observations supporting each event-month m under the baseline NYT specification.